

A Technical Introduction to CLIP neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

Section 1

Spiking neural networks (SNNs) are artificial neural networks (ANN) that mimic natural neural networks. These models leverage timing of discrete spikes as the main information carrier. In addition to neuronal and synaptic state, SNNs incorporate the concept of time into their operating model. The idea is that neurons in the SNN do not transmit information at each propagation cycle (as it happens with typical multi-layer perceptron networks), but rather transmit information only when a membrane potential—an intrinsic quality of the neuron related to its membrane electrical charge—reaches a specific value, called the threshold. When the membrane potential reaches the threshold, the neuron fires, and generates a signal that travels to other neurons which, in turn, increase or decrease their potentials in response to this signal. A neuron model that fires at the moment of threshold crossing is also called a spiking neuron model. While spike rates can be considered the analogue of the variable output of a traditional ANN, neurobiology research indicated that high speed processing cannot be performed solely through a rate-based scheme. For example humans can perform an image recognition task requiring no more than 10ms of processing time per neuron through the successive layers (going from the retina to the temporal lobe). This time window is too short for rate-based encoding. The precise spike timings in a small set of spiking neurons also has a higher information coding capacity compared with a rate-based approach. The most prominent spiking neuron model is the leaky integrate-and-fire model. In that model, the momentary activation level (modeled as a differential equation) is normally considered to be the neuron's state, with incoming spikes pushing this value higher or lower, until the state eventually either decays or—if the firing threshold is reached—the neuron fires. After firing, the state variable is reset to a lower value. Various decoding methods exist for interpreting the outgoing spike train as a real-value number, relying on either the frequency of spikes (rate-code), the time-to-first-spike after stimulation, or the interval between spikes.

Section 2

Information in the brain is represented as action potentials (neuron spikes), which may group into spike trains or coordinated waves. A fundamental question of neuroscience is to determine whether neurons

communicate by a rate or temporal code. Temporal coding implies that a single spiking neuron can replace hundreds of hidden units on a conventional neural net. SNNs define a neuron's current state as its potential (possibly modeled as a differential equation). An input pulse causes the potential to rise and then gradually decline. Encoding schemes can interpret these pulse sequences as a number, considering pulse frequency and pulse interval. Using the precise time of pulse occurrence, a neural network can consider more information and offer better computing properties. SNNs compute in the continuous domain. Such neurons test for activation only when their potentials reach a certain value. When a neuron is activated, it produces a signal that is passed to connected neurons, accordingly raising or lowering their potentials. The SNN approach produces a continuous output instead of the binary output of traditional ANNs. Pulse trains are not easily interpretable, hence the need for encoding schemes. However, a pulse train representation may be more suited for processing spatiotemporal data (or real-world sensory data classification). SNNs connect neurons only to nearby neurons so that they process input blocks separately (similar to CNN using filters). They consider time by encoding information as pulse trains so as not to lose information. This avoids the complexity of a recurrent neural network (RNN). Impulse neurons are more powerful computational units than traditional artificial neurons. SNNs are theoretically more powerful than so called "second-generation networks" defined as ANNs "based on computational units that apply activation function with a continuous set of possible output values to a weighted sum (or polynomial) of the inputs"; however, SNN training issues and hardware requirements limit their use. Although unsupervised biologically inspired learning methods are available such as Hebbian learning and STDP, no effective supervised training method is suitable for SNNs that can provide better performance than second-generation networks. Spike-based activation of SNNs is not differentiable, thus gradient descent-based backpropagation (BP) is not available. SNNs have much larger computational costs for simulating realistic neural models than traditional ANNs. Pulse-coupled neural networks (PCNN) are often confused with SNNs. A PCNN can be seen as a kind of SNN. Researchers are actively working on various topics. The first concerns differentiability. The expressions for both the forward- and backward-learning methods contain the derivative of the neural activation function which is not differentiable because a neuron's output is either 1 when it spikes, and 0 otherwise. This all-or-nothing behavior disrupts gradients and makes these neurons unsuitable for gradient-based optimization. Approaches to resolving it include:

Section 3

Hardware Efforts to implement hardware-based spiking neural networks (SNNs) began in the 1980s, when

researchers began exploring brain-inspired neuromorphic systems. In the following decades, advancements in semiconductor technologies enabled the development of several notable projects. One such project is SpiNNaker, developed at the University of Manchester, which utilizes millions of processing cores for large-scale simulation of spiking neurons. TrueNorth, developed by IBM, is one of the first commercial neuromorphic chips, designed for energy-efficient and parallel processing. Loihi, an Intel research chip, focuses on online learning and adaptability in neuromorphic modeling. In research contexts, platforms such as BrainScaleS, developed in Europe, integrate analog and digital circuitry to accelerate neural simulations. Neurogrid, from Stanford University, was designed to efficiently simulate biological neurons and synapses. Dynap-se models developed by iniLabs, are a family of low-power, event-based neuromorphic chip intended for use in robotics and Internet of Things (IoT) applications. In addition, hardware based on memristors and other emerging memory technologies is being explored for the implementation of SNNs, with the goal of achieving lower power consumption and improved compatibility with biological neural models.

Section 4

Temporal coding; generating one spike per neuron, in which spike latency is inversely proportional to the pixel intensity. Rate coding; converting pixel intensity into a spike train, where the number of spikes is proportional to the pixel intensity. Direct coding; using a trainable layer to generate a floating-point value for each time step. The layer converts each pixel at a certain time step into a floating-point value, and then a threshold is used on the generated floating-point values to pick either zero or one. Phase coding; encoding temporal information into spike patterns based on a global oscillator. Burst coding; transmitting spikes in bursts, increasing communication reliability.